Suvadip Sana

PhD Candidate, Department of Statistics and Data Science, Cornell University
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Research Interests

I develop the statistical and algorithmic foundations for AI systems that must operate under diverse, often conflicting human preferences. My work spans **AI value alignment**, **Computational Social Choice**, **preference learning**, **democratic AI**, and **behavioral evaluation of language models**, drawing on statistical theory and social choice. Broader interests include **AI safety** and the **societal impacts of AI**.

Education

2021 – 2027 (exp.) **Cornell University**, Ithaca, NY, USA
PhD in Statistics
Advisor: Martin T. Wells , Lionel Levine

2021 – 2024 **Cornell University**, Ithaca, NY, USA
MS in Statistics GPA: 3.93 / 4.00

2019 – 2021 **Indian Statistical Institute**, Bengaluru, India
Master of Mathematics (M.Math)
First Division with Distinction

2016 – 2019 **Indian Statistical Institute**, Bengaluru, India
Bachelor of Mathematics (B.Math)
First Division with Distinction

Publications and Preprints

2026 **EigenBench: A Comparative Behavioral Measure of Value Alignment**
J. Chang, L. Piff, **Sana, S.**, J. X. Li, L. Levine
International Conference on Learning Representations (ICLR), 2026
Oral presentation (∼ top 1% of submissions) · Paper

2026 **Pluralistic Preference Alignment via Sortition-Weighted RLHF**
Sana, S., J. Wu, M. T. Wells
ICML 2026 Workshop on Pluralistic Alignment · Paper

2026 **Gradating Arrow's Axioms**
Sana, S., P. Rock, M. T. Wells, M. Duchin
Under Review · Preprint

2026 **Mixing Times of Glauber Dynamics on Masked Language Models**
Sana, S., S. Wolf, N. Mehta, A. Shah, A. Shaikh, J. Goodman, L. Levine
Under Review · arXiv:2605.16378

2026 **CivicAlign AI: Constitution-Guided Language Model Agents for Public-Sector Governance**
Sana, S., L. Levine, M. T. Wells
Working Paper · Draft

2026 **Statistical Aspects of Sortition**
Sana, S., M. T. Wells · *In preparation*

2026 **AI Value Evolution**
Sana, S., L. Levine · *In preparation*

Technical Reports

2025 **Quantitative Relaxations of Arrow's Axioms**
Sana, S., D. Brous, M. T. Wells, M. Duchin
Preliminary version of *Gradation of Arrow's Axioms* · arXiv:2506.12961

2024 **A Scalar Matching Factor on the Birkhoff Polytope Characterizing Permutation and Uniform Matrices**
Sana, S. · arXiv:2406.16284

Research and Internship Experience

Fall 2026 **AI Engineering Intern**, New York City Council Data Team
Susan and Rubinstein PiTech Innovation Fund, Cornell Tech · Advisor: Alaa Moussawi
Optimizing the retrieval-augmented generation (RAG) system used by NYC Council staff for legislative research.

Summer 2026 **PiTech Impact Fellow (AI Engineering)**, New York City Council Data Team
Advisor: Alaa Moussawi
Built an AI routing system and MCP server powering a staff-facing chatbot that directs queries across multiple generative-AI applications and external tools, giving NYC Council staff faster, more accurate answers to complex legislative questions through a single interface.

Summer 2025 **Summer Fellow**, Data and Democracy Lab, Cornell Brooks School of Public Policy
Advisor: Moon Duchin
Developed statistical fairness measures for ranked-choice voting systems, enabling empirical evaluation of how closely electoral outcomes align with principles from social choice theory.

Apr – Jun 2020 **Statistical Analyst Intern**, Siemens Healthineers, Bengaluru, India

Awards and Honors

2026 Susan and Rubinstein Innovation Fund, Cornell Tech

2026 ICML 2026 Silver Reviewer Award

2026 Cornell Bowers CIS Social Impact and Service Award

2026 Cornell PiTech Impact Fellowship

2025 Tapia Conference Travel Award *($1,500)*

2024 NSF Travel Award *($500, IISA 2024)*

2024 Cornell Graduate School Research Travel Grant *($1,300, study of ranked-choice elections)*

2021 – 2022 Cornell PhD Graduate Fellowship *(awarded to two incoming PhD students)*

2020 NBHM Master's Scholarship, Government of India

2019 All India Rank 3, M.Stat entrance examination, Indian Statistical Institute

2019 All India Rank 33, IIT JAM (Mathematical Statistics)

2016 – 2019 ISI Teachers' Award, Indian Statistical Institute *(awarded to top 3 students)*

Invited Talks and Presentations

Dec 2026 New York City Technology Forum, Brooklyn, NYC *(Upcoming invited speaker and panelist)*

Nov 2026 Cornell Math Department Probability Seminar, Cornell University *(upcoming invited talk)*

Oct 2026 PiTech showcase , Cornell Tech, NYC *(Invited talk)*

Feb 2026 Social Choice for AI Ethics and Safety Workshop, IASEAI 2026, Paris, France *(invited talk)*

Jan 2026 Alignment Research Center, Berkeley, CA, USA *(invited talk)*

Oct 2025 Social Choice: Theory and Computation, Wellesley College, MA, USA *(poster)*

Jul 2025 Fairness in Social Choice Workshop, Northwestern University, IL, USA *(invited talk)*

Jan 2025 Joint Mathematics Meetings, Seattle, WA, USA *(contributed talk)*

Dec 2024 IISA 2024, Kerala, India *(contributed talk)*

Sep 2024 Statistics Graduate Student Seminar, Cornell University *(invited talk)*

Aug 2022 Hyundai Vision Conference, Seoul, South Korea *(poster)*

Teaching

Spring 2026 **Co-Instructor**, MATH 1710: Statistical Theory and Application in the Real World, Cornell University

Summer 2025 **Lecturer**, Research Practices and Academic Culture at BURE Undergraduate research seminar, Cornell University

Teaching Assistant, Cornell University

Spring 2026 MATH 2940: Linear Algebra for Engineers — weekly discussion sections and office hours

Fall 2023 – Fall 2025 MATH 6710: Probability Theory I (three semesters: Fall 2023, 2024, 2025) — office hours and grading

Spring 2025 MATH 6720: Probability Theory II — office hours and grading

Spring 2025 STSCI 4100: Multivariate Statistics — grading and lab sessions

Fall 2023 CS 6386: Data to Decisions — office hours

2023 STSCI 2100: Introductory Statistics (Winter and Summer sessions) — grading and office hours

Spring 2023 MATH 4720: Statistics — office hours and grading

Spring 2023 STSCI 5640: Statistics for Financial Engineering — grading

Fall 2022 STSCI 4090: Theory of Statistics — office hours and grading

Mentorship and Service

Sep 2025 – Present **Research Lead**, Cornell AI Alignment Club

Dec 2024 **Session Chair**, IISA 2024

2022 **Professional Development Chair**, Statistics Graduate Society, Cornell University

Spring 2022 **Graduate Ambassador**, Cornell CIS DEI

Academic Reviewing

2026 – ICML 2026 · NeurIPS 2026 · ICML 2026 Pluralistic Alignment Workshop · AAAI 2027 · ICLR 2027

Skills

Programming: Python, R (proficient); MATLAB (familiar)

Tools and frameworks: PyTorch, Hugging Face Transformers, RAG pipelines, MCP servers, FastAPI, Git, RunPod, VoteKit

Languages: Bengali (native), English, Hindi, Korean

Other Interests

Cooking, running, and traveling.

References

Martin T. Wells
Professor of Statistics
Cornell University
Email: mtw1@cornell.edu

Lionel Levine
Professor of Mathematics
Cornell University
Email: levine@math.cornell.edu

Moon Duchin
Professor of Mathematics
University of Chicago
Email: mduchin@uchicago.edu